Multiple Linear Regression

Multiple inputs and comparable feature scales

Worked answers

Name _____ Date _____

01 Data and Features

* Synthetic practice data Area is entered in units of 10 m² · For 120 m², use $x_1 = 12$

House i	Area (m²)	Input x_1	Rooms x_2	Target y (10M KRW)
1	120	12	3	21
2	60	6	2	12
3	160	16	4	28
4	100	10	3	22
5	180	18	4	31
6	80	8	2	15
7	140	14	3	25

1 Write the number of examples m and features n

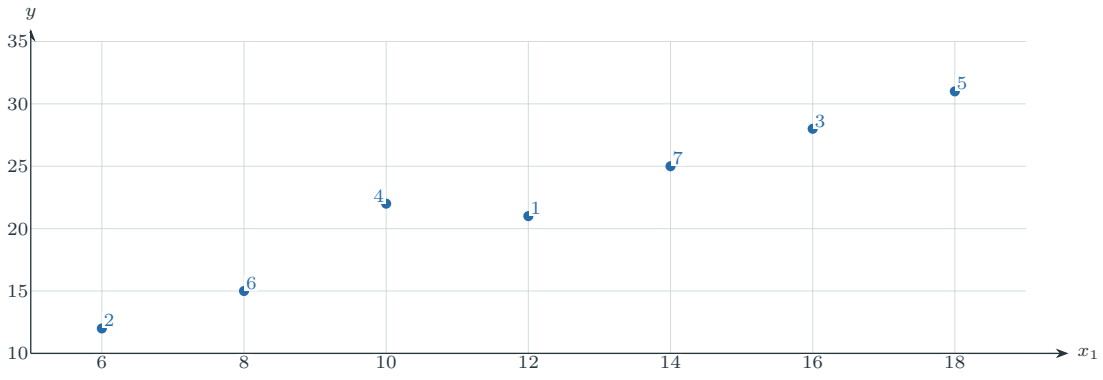
$m =$ 7 $n =$ 2

2 Express the input of House 4 as a vector and find the following values

$\mathbf{x}^{(i)} = [x_1^{(i)}, x_2^{(i)}]$ i = house index, j = feature index

$\mathbf{x}^{(4)} = [$ 10 , 3 $]$ $x_2^{(4)} =$ 3 $y^{(4)} =$ 22

3 Is a larger house always more expensive in the actual data? Compare Houses 1 and 4



☐ Always more expensive ☐ Not always

House 4: 100 m², KRW 220 million House 1: 120 m², KRW 210 million
Larger houses tend to cost more, but area alone cannot explain every price difference

Reference · DeepLearning.AI / Stanford Online · Machine Learning Specialization · Current lesson slides 83 to 99

02 Prediction and Cost

$f_{\mathbf{w},b}(\mathbf{x}) = w_1x_1 + w_2x_2 + b \qquad \mathbf{w} = [1, 3], \quad b = 2$

4 Calculate the price of House 4 using both the expanded expression and a dot product

$\mathbf{w} \cdot \mathbf{x} = w_1x_1 + w_2x_2 \qquad f_{\mathbf{w},b}(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} + b$

$1 \times 10 + 3 \times 3 + 2 = \quad 10 + 9 + 2 = 21$

$[1, 3] \cdot [10, 3] + 2 = \quad 19 + 2 = 21$

A dot product multiplies matching entries and adds the products Add b once at the end
Two inputs still produce one predicted price

5 Calculate each of the seven predictions, errors, and squared errors

$\hat{y}^{(i)} = x_1^{(i)} + 3x_2^{(i)} + 2 \qquad e^{(i)} = \hat{y}^{(i)} - y^{(i)}$

House	Prediction expression	$y^{(i)}$	$\hat{y}^{(i)}$	$e^{(i)}$	$(e^{(i)})^2$
1	$12 + 3 \times 3 + 2$	21	$12 + 9 + 2 = 23$	$23 - 21 = 2$	$(2)^2 = 4$
2	$6 + 3 \times 2 + 2$	12	$6 + 6 + 2 = 14$	$14 - 12 = 2$	$(2)^2 = 4$
3	$16 + 3 \times 4 + 2$	28	$16 + 12 + 2 = 30$	$30 - 28 = 2$	$(2)^2 = 4$
4	$10 + 3 \times 3 + 2$	22	$10 + 9 + 2 = 21$	$21 - 22 = -1$	$(-1)^2 = 1$
5	$18 + 3 \times 4 + 2$	31	$18 + 12 + 2 = 32$	$32 - 31 = 1$	$(1)^2 = 1$
6	$8 + 3 \times 2 + 2$	15	$8 + 6 + 2 = 16$	$16 - 15 = 1$	$(1)^2 = 1$
7	$14 + 3 \times 3 + 2$	25	$14 + 9 + 2 = 25$	$25 - 25 = 0$	$(0)^2 = 0$

6 Calculate the MSE and cost

$\text{MSE} = \frac{1}{m}\sum_{i=1}^{m}(e^{(i)})^2 \qquad J(\mathbf{w}, b) = \frac{1}{2m}\sum_{i=1}^{m}(e^{(i)})^2 = \frac{1}{2}\text{MSE}$

$\sum_{i=1}^{7}(e^{(i)})^2 = \qquad 4 + 4 + 4 + 1 + 1 + 1 + 0 = 15$

$\text{MSE} = \qquad \frac{15}{7} \qquad J(\mathbf{w}, b) = \qquad \frac{15}{14}$

MSE $= 15 \div 7 = \frac{15}{7}$ $J = 15 \div (2 \times 7) = \frac{15}{14} \approx 1.071$
m counts houses Even with two features, divide by $2m = 14$

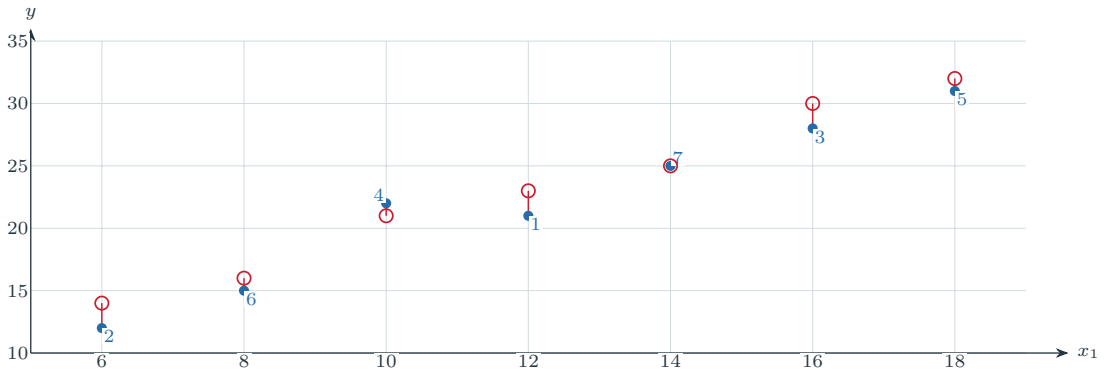
03 Prediction Plot

7 Use each house's area and room count to plot **the predictions from Question 5**
Draw circles, then connect each prediction to the same house's actual price with a **vertical line**

$P_i = (x_1^{(i)}, \hat{y}^{(i)})$ $\hat{y}^{(i)} = x_1^{(i)} + 3x_2^{(i)} + 2$

$i=1$	$i=2$	$i=3$	$i=4$	$i=5$	$i=6$	$i=7$
$(12, 23)$	$(6, 14)$	$(16, 30)$	$(10, 21)$	$(18, 32)$	$(8, 16)$	$(14, 25)$

* Same axes and actual data as Question 3 Blue points: actual prices · Circles: predictions



Use the plot to check the error signs for Houses 1 and 4

$e^{(i)} = \hat{y}^{(i)} - y^{(i)}$ vertical distance $= |e^{(i)}|$

$e^{(1)} =$ $23 - 21 = 2$ $e^{(4)} =$ $21 - 22 = -1$

House 1: prediction is 2 above the actual price; its circle is above the blue point
House 4: prediction is 1 below the actual price; its circle is below the blue point
House 7: $25 - 25 = 0$, so the points overlap
A longer vertical line means a larger error magnitude; the cost uses its square

04 Parameter Update

$\mathbf{w} = [1, 3], \quad b = 2 \qquad \alpha = 0.01 \qquad m = 7$

8 Copy the errors from Question 5 and multiply each by the corresponding feature

$e^{(i)} = f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)} \qquad e^{(i)}x_1^{(i)}, \quad e^{(i)}x_2^{(i)}$

House	$e^{(i)}$	$e^{(i)}x_1^{(i)}$	$e^{(i)}x_2^{(i)}$
1	2	$2 \times 12 = 24$	$2 \times 3 = 6$
2	2	$2 \times 6 = 12$	$2 \times 2 = 4$
3	2	$2 \times 16 = 32$	$2 \times 4 = 8$
4	-1	$-1 \times 10 = -10$	$-1 \times 3 = -3$
5	1	$1 \times 18 = 18$	$1 \times 4 = 4$
6	1	$1 \times 8 = 8$	$1 \times 2 = 2$
7	0	$0 \times 14 = 0$	$0 \times 3 = 0$
Sum	7	84	21

9 Calculate the gradient for each parameter

$$g_1 = \frac{\partial J}{\partial w_1} = \frac{1}{m}\sum_{i=1}^{m} e^{(i)}x_1^{(i)} \qquad g_2 = \frac{\partial J}{\partial w_2} = \frac{1}{m}\sum_{i=1}^{m} e^{(i)}x_2^{(i)} \qquad g_b = \frac{\partial J}{\partial b} = \frac{1}{m}\sum_{i=1}^{m} e^{(i)}$$

$g_1 = \frac{84}{7} = 12 \qquad g_2 = \frac{21}{7} = 3 \qquad g_b = \frac{7}{7} = 1$

* Use the provided derivatives The gradient for b has no feature multiplier

10 Update all three values simultaneously from the same starting point

$w_1^{\text{new}} = w_1 - \alpha g_1 \qquad w_2^{\text{new}} = w_2 - \alpha g_2 \qquad b^{\text{new}} = b - \alpha g_b$

$w_1^{\text{new}} = 1 - 0.01 \times \quad 12 \quad = \quad 0.88$

$w_2^{\text{new}} = 3 - 0.01 \times \quad 3 \quad = \quad 2.97 \qquad b^{\text{new}} = 2 - 0.01 \times \quad 1 \quad = \quad 1.99$

$0.01 \times 12 = 0.12$, so $w_1^{\text{new}} = 1 - 0.12 = 0.88$
$w_2^{\text{new}} = 3 - 0.03 = 2.97 \quad b^{\text{new}} = 2 - 0.01 = 1.99$

11 Can you update w_1 first, then use its new value to recalculate g_2?

☐ Yes ☐ No

Calculate g_1, g_2, g_b using the same pre-update $\mathbf{w} = [1, 3], b = 2$
Use the same seven errors, then apply all new parameter values together

05 Cost and Feature Scale

12 Calculate House 4's error after the update and compare it with the overall cost change

$f_{\text{new}}(\mathbf{x}) = 0.88x_1 + 2.97x_2 + 1.99 \qquad \mathbf{x}^{(4)} = [10, 3], \quad y^{(4)} = 22$

$\hat{y}^{(4)}_{\text{new}} = \quad 8.8 + 8.91 + 1.99 = 19.70 \qquad e^{(4)}_{\text{new}} = 19.70 - 22 = -2.30$

* Overall cost across all seven houses: $J_{\text{old}} = 15/14 \approx 1.071$, $J_{\text{new}} = 11.8016/14 \approx 0.843$

☐ Lower cost means every error decreases ☐ Some errors can increase

House 4's error magnitude grows from $|21 - 22| = 1$ to $|19.70 - 22| = 2.30$
Overall J drops from about 1.071 to 0.843: it evaluates all seven errors together

13 For House 1, compare prediction changes when each weight is increased by 0.1 separately

$\Delta\hat{y} = x_j \Delta w_j \qquad \mathbf{x}^{(1)} = [12, 3] \quad (\text{other values fixed})$

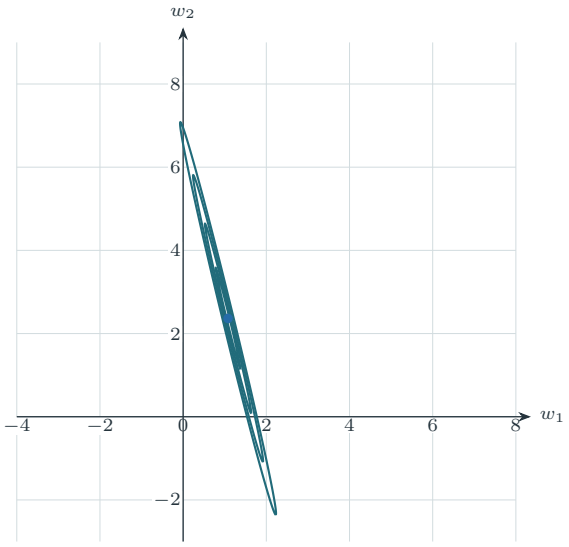
$\Delta w_1 = 0.1 \Rightarrow \Delta\hat{y} = \quad 12 \times 0.1 = 1.2 \qquad \Delta w_2 = 0.1 \Rightarrow \Delta\hat{y} = \quad 3 \times 0.1 = 0.3$

The same 0.1 weight change gives a larger prediction change for area
Weight magnitude alone does not establish feature importance; input units and scale also matter

14 The contours of our cost function are shown below

Analyze any features that could make it difficult for gradient descent to reach the minimum

* Same seven examples, $J(w_1, w_2, b = 2)$ A slice at fixed $b = 2$ · Blue point: minimum of this slice



Very elongated contours imply that the cost changes at very different rates in different directions
Large steps can cross the narrow valley to the opposite side
A small learning rate can make progress along the valley slow, requiring more iterations

06 Feature Scaling

15 Calculate the reference values for each feature

$$\mu_j = \frac{1}{m}\sum_{i=1}^{m} x_j^{(i)} \qquad R_j = x_{j,\text{max}} - x_{j,\text{min}}$$

Feature	Min.	Max.	Mean μ_j	Range R_j
x_1	6	18	12	12
x_2	2	4	3	2

$\mu_1 = (12+6+16+10+18+8+14)/7 = 84/7 = 12 \quad R_1 = 18-6 = 12$
$\mu_2 = (3+2+4+3+4+2+3)/7 = 21/7 = 3 \quad R_2 = 4-2 = 2$

16 Divide each feature by its own maximum

$$x_{1,\text{scaled}} = \frac{x_1}{18} \qquad x_{2,\text{scaled}} = \frac{x_2}{4}$$

House	$x_1/18$	$x_2/4$
1	$\frac{12}{18} = \frac{2}{3}$	$\frac{3}{4} = 0.75$
2	$\frac{6}{18} = \frac{1}{3}$	$\frac{2}{4} = 0.5$
3	$\frac{16}{18} = \frac{8}{9}$	$\frac{4}{4} = 1$
4	$\frac{10}{18} = \frac{5}{9}$	$\frac{3}{4} = 0.75$
5	$\frac{18}{18} = 1$	$\frac{4}{4} = 1$
6	$\frac{8}{18} = \frac{4}{9}$	$\frac{2}{4} = 0.5$
7	$\frac{14}{18} = \frac{7}{9}$	$\frac{3}{4} = 0.75$

17 Apply mean normalization

$$x_{j,\text{mean}} = \frac{x_j - \mu_j}{x_{j,\text{max}} - x_{j,\text{min}}} \qquad x_{1,\text{mean}} = \frac{x_1 - 12}{12} \qquad x_{2,\text{mean}} = \frac{x_2 - 3}{2}$$

House	$(x_1-12)/12$	$(x_2-3)/2$
1	$\frac{12-12}{12} = 0$	$\frac{3-3}{2} = 0$
2	$\frac{6-12}{12} = -0.5$	$\frac{2-3}{2} = -0.5$
3	$\frac{16-12}{12} = \frac{1}{3}$	$\frac{4-3}{2} = 0.5$
4	$\frac{10-12}{12} = -\frac{1}{6}$	$\frac{3-3}{2} = 0$
5	$\frac{18-12}{12} = 0.5$	$\frac{4-3}{2} = 0.5$
6	$\frac{8-12}{12} = -\frac{1}{3}$	$\frac{2-3}{2} = -0.5$
7	$\frac{14-12}{12} = \frac{1}{6}$	$\frac{3-3}{2} = 0$

18 What does a negative transformed room count mean?

☐ The house has a negative number of rooms ☐ Fewer rooms than the mean

For two rooms, $(2-3)/2 = -0.5$: fewer than the mean of three
Subtracting the mean centers at 0; dividing by the range reduces numerical scale

07 Z-score and New Inputs

19 Apply z-score normalization

$$x_{j,z} = \frac{x_j - \mu_j}{\sigma_j} \qquad \mu_1 = 12, \quad \sigma_1 = 4 \qquad \mu_2 = 3, \quad \sigma_2 = \frac{2}{\sqrt{7}} \approx 0.756$$

* Standard deviation σ measures spread around the mean; use the given values
Use fractions or three decimal places For room count, you may use $1/\sigma_2 \approx 1.323$

House	$(x_1 - 12)/4$	$(x_2 - 3)/(2/\sqrt{7})$
1	$\frac{12-12}{4} = 0$	$(3-3) \times \frac{\sqrt{7}}{2} = 0$
2	$\frac{6-12}{4} = -1.5$	$(2-3) \times \frac{\sqrt{7}}{2} = -\frac{\sqrt{7}}{2} \approx -1.323$
3	$\frac{16-12}{4} = 1$	$(4-3) \times \frac{\sqrt{7}}{2} = \frac{\sqrt{7}}{2} \approx 1.323$
4	$\frac{10-12}{4} = -0.5$	$(3-3) \times \frac{\sqrt{7}}{2} = 0$
5	$\frac{18-12}{4} = 1.5$	$(4-3) \times \frac{\sqrt{7}}{2} = \frac{\sqrt{7}}{2} \approx 1.323$
6	$\frac{8-12}{4} = -1$	$(2-3) \times \frac{\sqrt{7}}{2} = -\frac{\sqrt{7}}{2} \approx -1.323$
7	$\frac{14-12}{4} = 0.5$	$(3-3) \times \frac{\sqrt{7}}{2} = 0$

House 5 area: $(18 - 12)/4 = 1.5$, or 1.5 standard deviations above the mean
Four rooms give $(4 - 3)/(2/\sqrt{7}) = \sqrt{7}/2 \approx 1.323$; values need not lie between -1 and 1

20 Check the equations that preserve predictions when input units change

$$u_1 = \frac{x_1}{18}, \quad u_2 = \frac{x_2}{4} \qquad x_1 + 3x_2 + 2 = 18u_1 + 12u_2 + 2$$

* The same model in different units; keeping the old weights after scaling can change predictions

Use a new house of 150 m^2 with 3 rooms $\mathbf{x} = [15, 3]$

$u_1 =$ $\frac{15}{18} = \frac{5}{6}$ $u_2 =$ $\frac{3}{4}$

$15 + 3 \times 3 + 2 =$ 26 $18u_1 + 12u_2 + 2 =$ $15 + 9 + 2 = 26$

21 Transform a new house of 210 m^2 with 5 rooms using the same reference values

$$u_1 = \frac{21}{18} \qquad u_2 = \frac{5}{4} \qquad \text{use training reference values } 18, 4$$

$u_1 =$ $\frac{7}{6}$ $u_2 =$ 1.25

Do not replace the reference values with 21 and 5 for the new house; keep the training scale
New inputs can exceed the training range; transformed values above 1 are valid

22 Choose the correct statement

☐ Z-scores always lie between -1 and 1 ☐ Each feature uses its own mean and standard deviation

Area: $\mu_1 = 12, \sigma_1 = 4$ Rooms: $\mu_2 = 3, \sigma_2 = 2/\sqrt{7}$
The goal is roughly comparable numerical scales, not exactly filling a fixed interval

08 Scaling and Contours

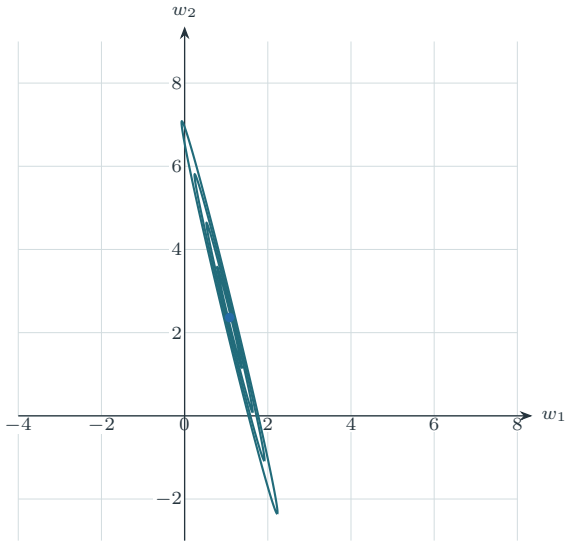
23 Compare the contours from Question 14 with those after scaling Explain any improvement

$$u_1 = \frac{x_1}{18}, \quad u_2 = \frac{x_2}{4} \qquad v_1 = 18w_1, \quad v_2 = 4w_2 \qquad b = 2$$

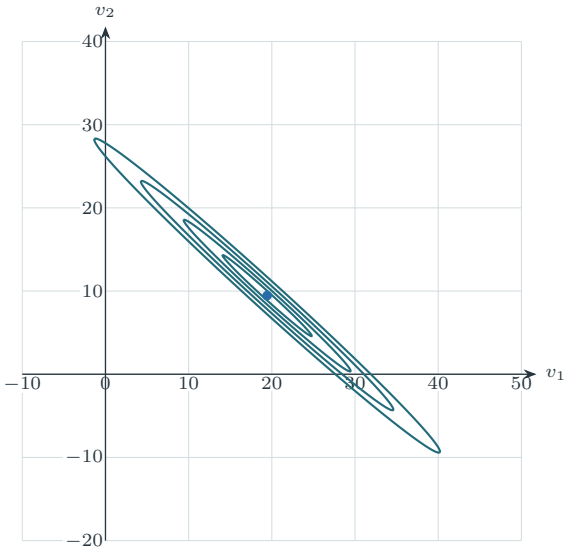
$$J_{\text{scaled}}(v_1, v_2) = \frac{1}{2 \times 7} \sum_{i=1}^{7} \left(v_1 u_1^{(i)} + v_2 u_2^{(i)} + 2 - y^{(i)} \right)^2$$

* Same houses and targets as Question 14; fixed $b = 2$ Both axes use equal scale within each plot

Q14 / Before



Q16 / After



* Both plots show costs $0.1,\ 0.35,\ 0.8,\ 1.5$ above the minimum

The contours are less elongated, so directional differences in cost change are reduced
With an appropriate learning rate, updates can cross the valley less sharply and reach the minimum more efficiently
Area and room count remain correlated, so the contours remain elliptical
Equivalent prediction weights give the same cost; scaling alone has not reduced prediction error